Trigonometric Substitution (7.3)

$$\int \sqrt{16-x^2}\,dx$$

???

$$\int x\sqrt{16-x^2}\,dx$$

$$u = 16 - x^2$$

$$du = -2x\,dx$$

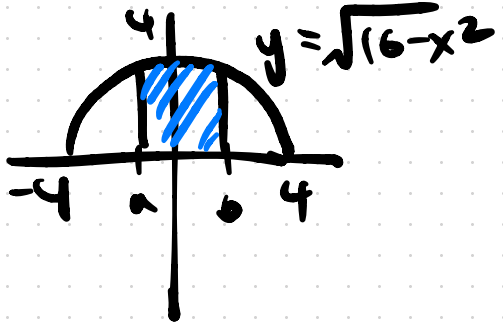
$$-\frac{1}{2}du = x\,dx$$

$$\int \sqrt{u}\left(-\frac{1}{2}du\right)$$

$$\int_a^b \sqrt{16-x^2}\,dx$$

$$x^2+y^2=16$$

$$y=\sqrt{16-x^2}$$

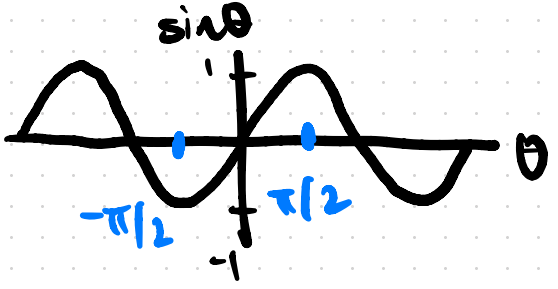


$$\int \frac{1}{\sqrt{1-x^2}}\,dx$$

Introduce a variable θ. We want $x = \sin\theta$.

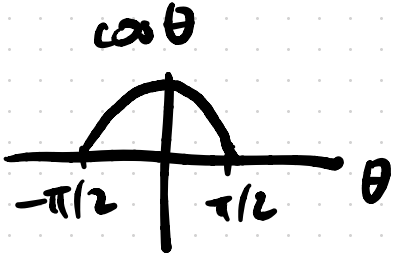
This is okay because the function in the integral is only defined for $-1 \le x \le 1$

Also, $\sin\theta$ can take any value between -1 and 1.



so for every possible x between -1 and 1, there is some θ such that $x = \sin\theta$. We can find such θ with $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$.

$$\int \frac{1}{\sqrt{1-x^2}}\,dx$$



Let $x = \sin\theta$, where $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

$$dx = \cos\theta\,d\theta$$

$$\int \frac{1}{\sqrt{1-\sin^2\theta}}\cos\theta\,d\theta$$

$$\int \frac{1}{\cos\theta}\cos\theta\,d\theta$$

$$\sqrt{1-\sin^2\theta} = \sqrt{\cos^2\theta}$$

$$= |\cos\theta|$$

$$= \cos\theta$$

because $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

$$\int d\theta$$

$$\theta + C$$

$$\sin^{-1} x + C$$

$$x = \sin\theta$$

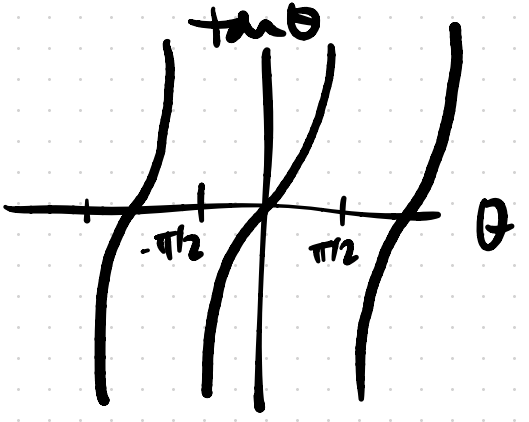
$$\theta = \sin^{-1} x$$

↑

principal branch of $\sin^{-1}$, i.e.

$$-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$$

$$\int \frac{1}{1+x^2}\,dx$$



Let $x = \tan\theta \qquad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$$dx = \sec^2\theta\,d\theta$$

$$1+x^2 = 1+\tan^2\theta = \sec^2\theta$$

$$\int \frac{1}{\sec^2\theta}\cdot \sec^2\theta\,d\theta$$

↙ principal branch, i.e. $-\frac{\pi}{2} < \tan^{-1}x < \frac{\pi}{2}$

$$\int d\theta = \theta + C = \tan^{-1}x + C$$

"Rules of thumb" for trigonometric substitution.

$\sqrt{a^2-x^2}$ $a>0$	$x = a\sin\theta \quad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ $\sqrt{a^2-x^2} = \sqrt{a^2-a^2\sin^2\theta} = a\sqrt{1-\sin^2\theta} = a\lvert\cos\theta\rvert = a\cos\theta$
$\sqrt{a^2+x^2}$ $a>0$	$x = a\tan\theta \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$ $\sqrt{a^2+x^2} = \sqrt{a^2+a^2\tan^2\theta} = a\sqrt{1+\tan^2\theta} = a\lvert\sec\theta\rvert = a\sec\theta$
$\sqrt{x^2-a^2}$ $a>0$	$x = a\sec\theta \quad 0 \le \theta < \frac{\pi}{2}$ or $\pi \le \theta < \frac{3\pi}{2}$ $\sqrt{x^2-a^2} = \sqrt{a^2\sec^2\theta-a^2} = a\sqrt{\sec^2\theta-1} = a\lvert\tan\theta\rvert$ $= a\tan\theta$

$$\int \sqrt{16-x^2}\,dx \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$$

Let $x = 4\sin\theta$ $\qquad \sqrt{16-x^2} = \sqrt{16-16\sin^2\theta} = 4\cos\theta$

$dx = 4\cos\theta\,d\theta$

$$\int 4\cos\theta \cdot 4\cos\theta\,d\theta$$

$$\int 16\cos^2\theta\,d\theta \qquad \cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$$

$$\int 8(1+\cos 2\theta)\,d\theta$$

$8\theta + 4\sin 2\theta + C$

$8\sin^{-1}\frac{x}{4}$

$8\sin\theta\cos\theta$

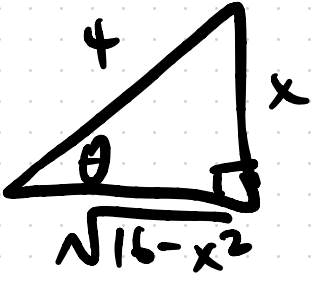
$8\sin\left(\sin^{-1}\frac{x}{4}\right)\cos\theta$

$8 \cdot \frac{x}{4} \qquad \frac{\sqrt{16-x^2}}{4}$

$x = 4\sin\theta$

$\sin\theta = \frac{x}{4}$

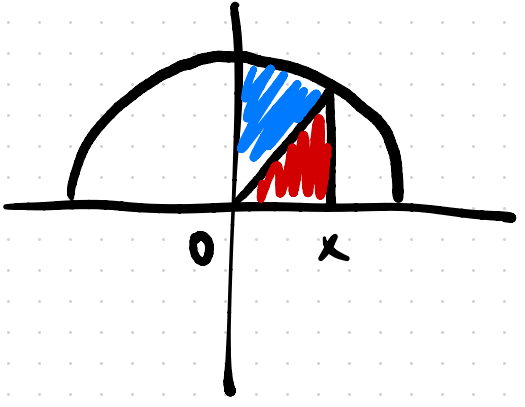
$\theta = \sin^{-1}\frac{x}{4}$



$\sin\theta = \frac{x}{4}$

$\cos\theta = \frac{\sqrt{16-x^2}}{4}$

$$8\sin^{-1}\frac{x}{4} + \frac{1}{2}x\sqrt{16-x^2} + C$$



$$\int_0^2 \frac{1}{(x^2+4)^{3/2}}\,dx$$

$$\int_0^2 \frac{1}{(\sqrt{x^2+4})^3}\,dx$$

Let $x = 2\tan\theta$

$dx = 2\sec^2\theta\,d\theta$

$$\int_0^{\pi/4} \frac{1}{(2\sec\theta)^3} \cdot 2\sec^2\theta\,d\theta$$

$$\int_{x=0}^{x=2}$$

$2\tan\theta = 2 \quad \tan\theta = 1 \quad \theta = \frac{\pi}{4}$

$2\tan\theta = 0 \quad \tan\theta = 0 \quad \theta = 0$

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$\sqrt{x^2+4} = \sqrt{4\tan^2\theta + 4}$$

$$= 2\sec\theta$$

$$\int_0^{\pi/4} \frac{1}{4} \cdot \frac{1}{\sec\theta} \, d\theta$$

$$\int_0^{\pi/4} \frac{1}{4} \cos\theta \, d\theta$$

$$\frac{1}{4} \sin\theta \Big|_0^{\pi/4} = \frac{1}{4} \frac{\sqrt{2}}{2}$$

$$\int_0^{2/3} \frac{1}{\sqrt{4-9x^2}}\,dx$$

$$\int_0^{2/3} \frac{1}{\sqrt{4-(3x)^2}}\,dx$$

Let $3x = 2\sin\theta$

$$x = \frac{2}{3}\sin\theta$$

$$dx = \frac{2}{3}\cos\theta\,d\theta$$

$$\sqrt{4-(3x)^2} = \sqrt{4-(2\sin\theta)^2}$$

$$= \sqrt{4-4\sin^2\theta}$$

$$= 2\cos\theta$$

$$\int_0^{\pi/2} \frac{1}{2\cos\theta} \cdot \frac{2}{3}\cos\theta \, d\theta$$

$x = \frac{2}{3}$ $\quad \frac{2}{3}\sin\theta = \frac{2}{3}$ $\quad \theta = \frac{\pi}{2}$

$x = 0$ $\quad \frac{2}{3}\sin\theta = 0$ $\quad \theta = 0$

$$\int_0^{\pi/2} \frac{1}{3} \, d\theta$$

$$\frac{1}{3} \cdot \frac{\pi}{2}$$

$$\int \frac{1}{\sqrt{x^2+x}}\,dx$$

$$\int \frac{1}{\sqrt{(x+\frac{1}{2})^2-\frac{1}{4}}}$$

Let $x+\frac{1}{2}=\frac{1}{2}\sec\theta$

$$\sqrt{(x+\tfrac{1}{2})^2-\tfrac{1}{4}}$$

$$\sqrt{(\tfrac{1}{2}\sec\theta)^2-\tfrac{1}{4}}$$

$$\sqrt{\tfrac{1}{4}\sec^2\theta-\tfrac{1}{4}}=\frac{1}{2}\tan\theta$$